

A 48 W Closed-Loop Buck Converter: Design and PSIM-Based Simulation Analysis

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Abstract

This study presents the design and closed-loop PSIM simulation analysis of a 48 W buck converter that steps down a 50 V DC input to a 24 V / 2 A output. During the design phase, the inductance, capacitance, and duty cycle parameters required for Continuous Conduction Mode (CCM) operation were theoretically calculated, and the voltage and current stresses on the semiconductor components were evaluated. To ensure system stability against sudden load transients and input fluctuations, a voltage-mode control architecture was implemented, utilizing a Type-III compensation network to optimize the frequency response. PSIM simulation results demonstrate that the proposed closed-loop controller precisely regulates the output voltage at the reference level, exhibits robust stability under transient conditions, and perfectly aligns with the theoretical calculations.

Keywords

Buck converter, closed-loop control, PSIM, Continuous Conduction Mode (CCM), Type-III compensation, DC-DC converter, voltage regulation.

1. Project Purpose

This project presents the design and closed-loop simulation of a 48 W buck (step-down) DC-DC converter in the PSIM environment, stepping down a 50 V DC input voltage to 24 V / 2 A. The objective of the study is not only to achieve the desired output voltage level but also to ensure system stability against sudden load changes, tolerate input voltage fluctuations, and maintain the inductor current in Continuous Conduction Mode (CCM).

2. Buck Converter and Open-Loop Limitations

The buck converter is one of the most fundamental power electronics topologies used to step down an input DC voltage to a lower DC voltage level. For an ideal buck converter, the duty cycle (D) is expressed as:

$$D = \frac{V_{out}}{V_{in}}$$
$$D = \frac{24}{50} = 0.48$$

This value indicates that the switch must remain in conduction for exactly 48% of each switching period. However, this approach is only valid under ideal conditions. In practical applications, conduction losses across the MOSFET and diode, the series resistance of the inductor, and load transients cause the output voltage of an open-loop system operating with a constant duty cycle to deviate from its reference value. Therefore, a closed-loop control architecture is required to maintain a constant output voltage across varying operating conditions.

2.1. Operating Principle of the Buck Converter

The buck (step-down) converter is a system that achieves a lower DC output voltage by switching the input voltage and passing it through an LC filter. The system is analyzed in two distinct operating periods based on the state of the switching device (MOSFET):

2.1.1. Conduction State (Switch ON / MOSFET ON)

When the PWM signal is active, the MOSFET enters conduction. In this state, current flows from the input source through the switch and the inductor to the load. The voltage across the inductor becomes $V_L = V_{in} - V_{out}$, and its current increases linearly as it stores energy in its magnetic field. The freewheeling

diode is reverse-biased and thus remains non-conducting.

2.1.2. OFF State (Switch OFF / MOSFET OFF)

When the PWM signal is inactive, the MOSFET turns off. Since the inductor current cannot change instantaneously, the freewheeling diode enters conduction, allowing the energy stored in the inductor to continue supplying the load. During this phase, the inductor current decreases linearly.

3. Why Closed-Loop Control?

In a closed-loop architecture, the output voltage is continuously monitored and compared with a reference voltage. The resulting error signal is processed through a compensation network to dynamically adjust the PWM duty cycle. Consequently, the output voltage is maintained at the reference level despite disturbances such as load increases or input voltage variations.

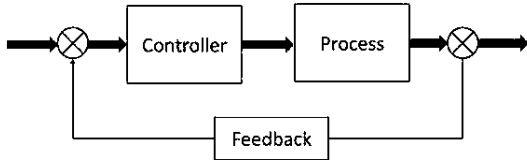


Figure 1. General control architecture.

4. Inductor and Current Ripple Design

Since the system is designed to operate in CCM, the inductor current ripple is selected as 25% of the maximum output current:

$$\Delta I_L = 2A \times 0.25 = 0.5A$$

The required inductance value for the buck converter is calculated using the following expression:

$$L = \frac{(V_{in} - V_{out})D}{\Delta I_L f_s}$$

$$L = \frac{(50 - 24) \times 0.48}{0.5 \times 250000} \approx 99.84 \mu H$$

Therefore, a standard value of **100 μH** , which is close to the calculated result, was selected for the simulation.

Assuming the average inductor current is equal to the output current:

$$I_{L,avg} = I_{out} = 2A$$

$$I_{L,max} = 2.25A, I_{L,min} = 1.75A$$

The fact that the minimum inductor current remains strictly above zero verifies that the system operates in CCM.

5. Output Capacitor (C_{out}) Calculation

The target for the output voltage ripple (Voltage Ripple, ΔV_{out}) is to be kept strictly below 1% (Max. **24V**). The minimum required output capacitance is calculated using the following expression:

$$C_{out} \geq \frac{\Delta I_L}{[8 * f_s * \Delta V_{out}]}$$

$$C_{out} \geq \frac{0.5}{[8 * 250000 * 0.24]} = 1.04 \mu F$$

However, in practical applications, the primary factor dictating the output voltage ripple is the Equivalent Series Resistance (ESR) of the capacitor:

$$\Delta V_{out(ESR)} = \Delta I_L * ESR$$

To ensure the voltage ripple does not exceed the 0.24 V limit, the ESR must be strictly less than **480m Ω** . To maintain a sufficient safety margin in both the design and simulation, as well as to stabilize the LC resonance frequency at a reliable point, an output capacitor of 10 μF was selected.

5.1. Semiconductor and Capacitor Stress (Derating) Analysis

For the physical hardware selection of the system, it is mandatory to calculate the average and RMS current values flowing through the components.

Input Capacitor RMS Current ($IC_{in,rms}$): Since the input current in buck converters is pulsed, the input capacitor is subjected to high RMS current stress. To prevent thermal damage to the capacitor, this value must be calculated:

$$IC_{in,rms} = I_{out} * \sqrt{[D * (1 - D)]}$$

$$IC_{in,rms} = 2A * \sqrt{[0.48 * 0.52]}$$

$$IC_{in,rms} = 2 * \sqrt{0.2496} = 2 * 0.499$$

$$= 0.998 A \text{ (Yaklaşık } 1.0 A \text{)}$$

Result: The rated ripple current of the selected input capacitor must be at least 1 A.

Output Capacitor RMS Current ($IC_{out,rms}$): Since the output capacitor filters the inductor current

ripple (ΔIL), the RMS stress on it is significantly lower compared to the input capacitor:

$$IC_{out_{rms}} = \Delta IL / \sqrt{12}$$

$$IC_{out_{rms}} = 0.5A / 3.464 = 0.144 A$$

Switching Device (MOSFET) Current and Voltage Stresses: The MOSFET carries the full load current during conduction and must withstand the entire input voltage when turned off.

Maximum Withstand Voltage (VDS_{max}) = V_{in} = 50 V

MOSFET RMS Current (IQ_{rms}) = $I_{out} * \sqrt{D}$

$$IQ_{rms} = 2A * \sqrt{0.48} = 2 * 0.692 = 1.38 A$$

Result: Considering the safety margin (derating), a MOSFET with a rating of at least 100 V and 3 A to 5 A should be selected.

Freewheeling Diode Stresses:

To determine the power dissipated across the diode, its average current is calculated:

Diode Average Current (ID_{avg}) = $I_{out} * (1 - D)$

$$ID_{avg} = 2A * (1 - 0.48) = 2 * 0.52 = 1.04 A$$

Maximum Reverse Voltage ($VRRM$) = V_{in} = 50 V

Result: To minimize conduction losses, a Schottky diode with a low forward voltage (V_f) and a rating of at least 100 V and 3 A must be used.

6. Feedback Scaling

A voltage divider is utilized to scale down the output voltage to the 0.8 V reference level used in the control circuit. For this purpose, $R_{top} = 290 k\Omega$ and $R_{bot} = 10 k\Omega$ were selected, thereby scaling the 24 V output voltage to a feedback level of approximately 0.8 V.

$$V_{out} = V_{ref} * \left(1 + \frac{R_{top}}{R_{bot}}\right)$$

7. Mathematics of the Type-III Compensation Network

In voltage-mode buck converters, the LC filter structure introduces a double-pole behavior, which increases phase lag and can adversely affect system stability. Therefore, a Type-III compensation

network is employed to attain an adequate phase margin and ensure the stable operation of the closed-loop system.

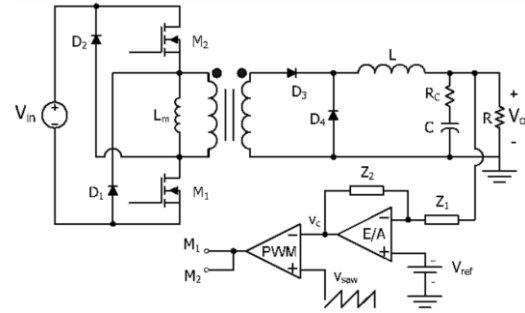


Figure 2. Type-III compensator with gain curve.

LC Resonance Frequency (Double Pole):

$$f_{LC} = \frac{1}{2\pi\sqrt{L * C_{out}}}$$

$$f_{LC} = \frac{1}{2\pi\sqrt{100\mu H * 100\mu F}} \approx 1.59 kHz$$

A Type-III compensation circuit, designed using a conventional Op-Amp, introduces an origin pole (f_{p0}), two zeros (f_{z1}, f_{z2}) and two high-frequency poles (f_{p1}, f_{p2}) to the system. With appropriate pole-zero placement, it can provide significant phase boost to the system.

Assuming $C_1 \gg C_3$ during the design phase, the pole and zero frequencies of the controller are mathematically expressed as follows:

Origin Pole:

$$f_{p0} = \frac{1}{[2\pi * R1 * C1]}$$

Zeros: Utilized to damp the phase drop caused by the LC filter.

$$f_{z1} = \frac{1}{2\pi * (R1 + R3) * C2}$$

$$f_{z2} = \frac{1}{2\pi * R2 * C2}$$

High-Frequency Poles: Employed to cancel the ESR zero and filter out high-frequency switching noise.

$$f_{p1} = \frac{1}{2\pi * R3 * C2}$$

$$f_{p2} \approx \frac{1}{2\pi * R2 * C3}$$

Once the required pole and zero frequencies of the system are determined, resistor R_1 is selected as a reference to calculate the resistance and capacitance values in the compensation circuit. All other compensation components are calculated using the following formulas:

Capacitor Calculations:

$$C_1 = \frac{f_{p2} - f_{z2}}{2\pi * R_1 * f_{p0} * f_{p2}}$$

$$C_2 = \frac{f_{p1} - f_{z1}}{2\pi * R_1 * f_{p1} * f_{z1}}$$

$$C_3 = \frac{f_{z2}}{2\pi * R_1 * f_{p0} * f_{z2}}$$

Direnç Hesaplamaları:

$$R_2 = \frac{R_1 * f_{p0} * f_{p2}}{(f_{p2} - f_{z2}) * f_{z2}}$$

$$R_3 = \frac{R_1 * f_{z1}}{(f_{p1} - f_{z1})}$$

8. Simulation Results and Verification

The findings obtained from the PSIM simulation of the closed-loop buck converter circuit demonstrate that the implemented Type-III compensation network stably regulates the system. The primary results obtained are summarized below:

Output Voltage and Current (Regulation): The average output voltage was obtained as approximately 24 V, and the average output current as approximately 2 A. Through closed-loop control, the system successfully locks onto the targeted 48 W power with a negligible deviation of 0.05%.

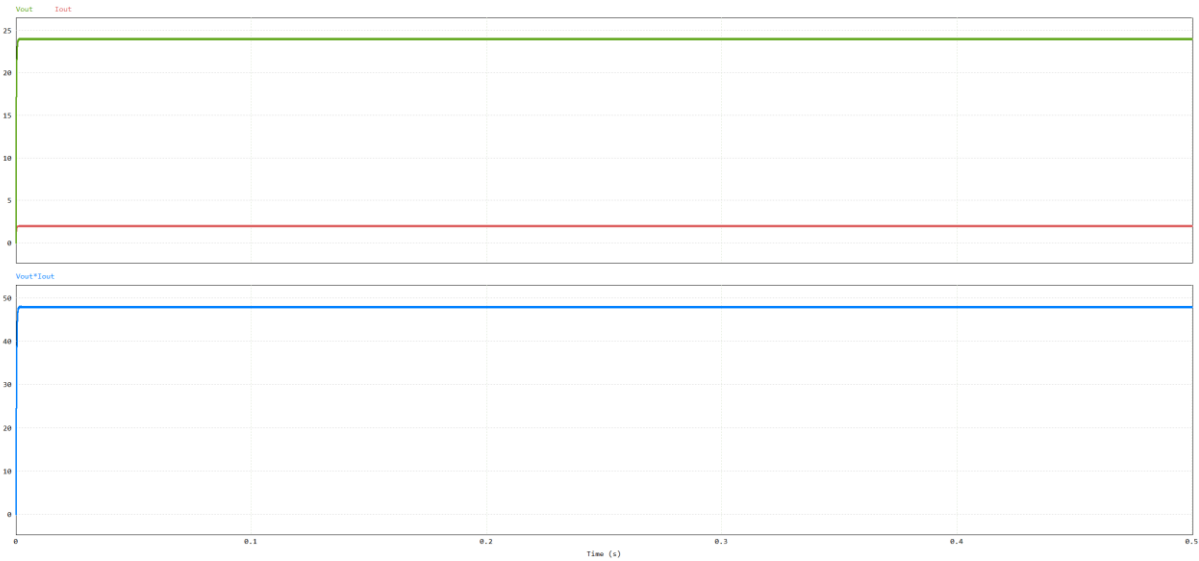


Figure 3. Output voltage, output current, and output power waveforms.

Inductor Current and Operating Mode (CCM): The average current flowing through the inductor is exactly 2.0 A. The peak current ($I_{L,max}$) was measured at 2.187 A, while the valley current ($I_{L,min}$) was recorded as 1.812 A. The fact that the minimum current remains significantly above zero proves that the circuit operates strictly in Continuous Conduction Mode (CCM).

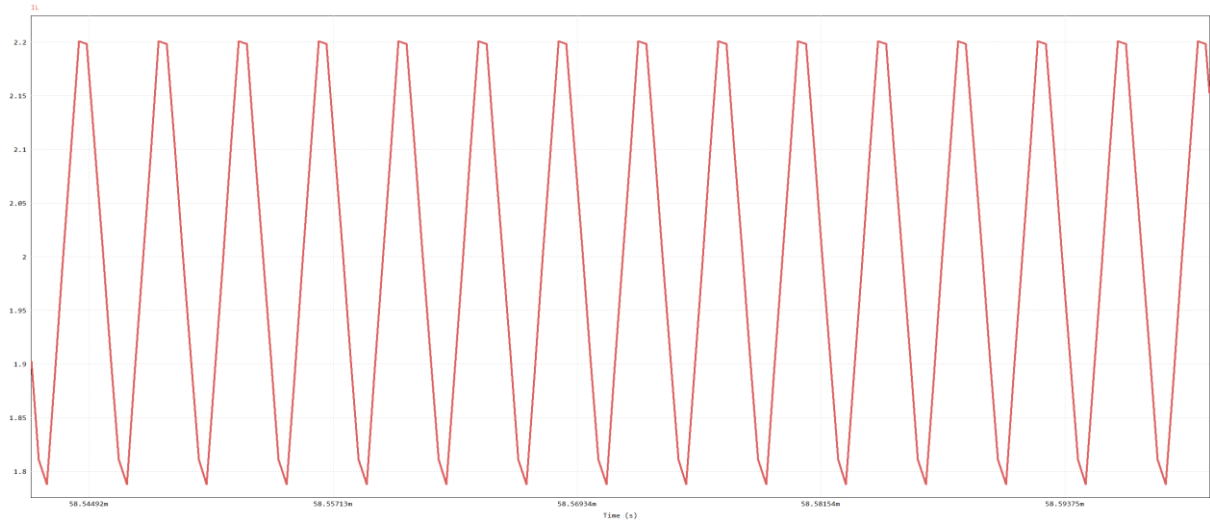


Figure 4. Inductor current waveforms.

Current Ripple Improvement: The inductor current ripple, which was theoretically targeted at 0.5 A, was observed to remain at approximately 0.375 A in the simulation. This result can be attributed to the realistic passive component models utilized in the circuit and the effects of closed-loop regulation.

Capacitor RMS Stress Verification: The RMS current of the output capacitor was measured at 140.9 mA (Theoretical: 144.0 mA), while the RMS current of the input capacitor was measured at 1.0 A (Theoretical: 0.998 A).

Measure							
	X1	X2	Δ	Average	Average X	RMS	
Time	1.00000e-01	4.00000e-01	3.00000e-01				
Iout	1.99907e+00	1.99846e+00	-6.11233e-04	1.99876e+00	1.99876e+00	1.99876e+00	
Vout	2.39888e+01	2.39815e+01	-7.33479e-03	2.39851e+01	2.39851e+01	2.39851e+01	
IL	2.18033e+00	2.11845e+00	-6.18815e-02	1.99953e+00	1.99953e+00	2.00454e+00	
IC	1.81182e-01	1.19913e-01	-6.12697e-02	6.90578e-04	1.19538e-01	1.40970e-01	

Figure 5. Measurement results.

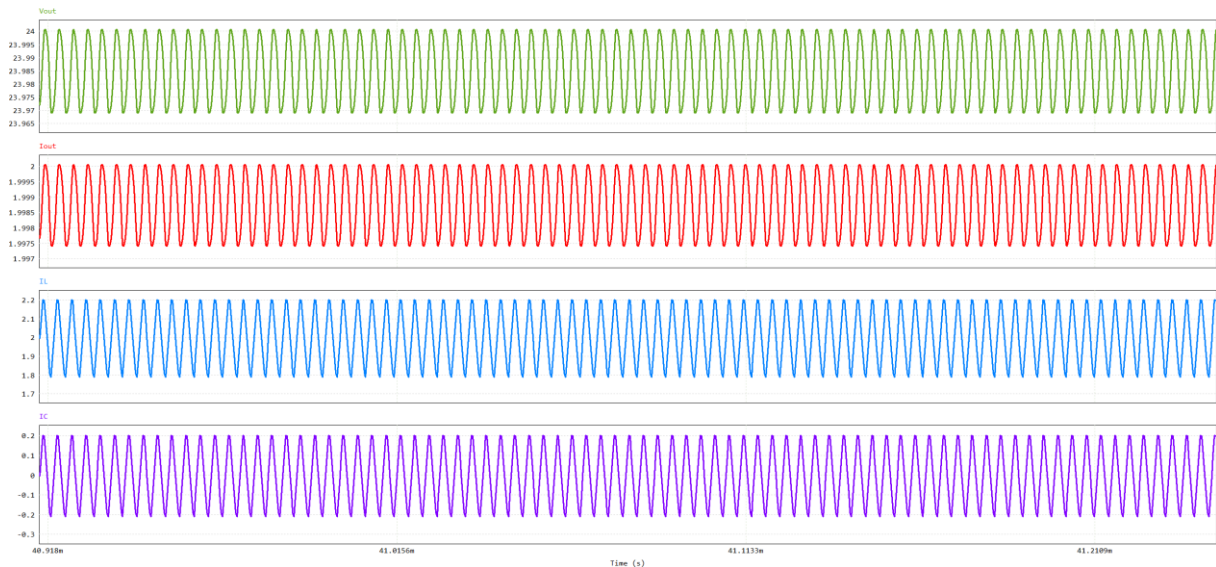


Figure 6. Output voltage, output current, inductor current, and capacitor current waveforms.

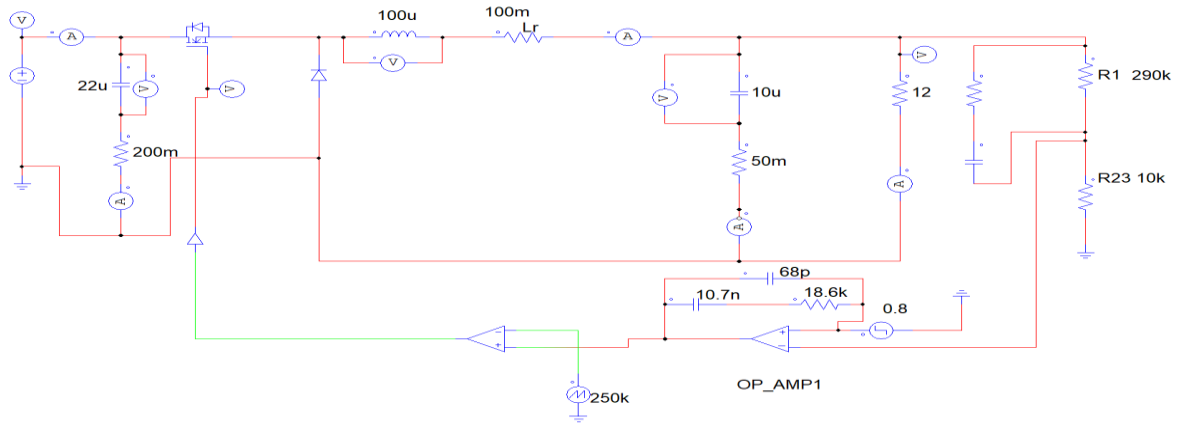


Figure 7. Circuit schematic.

9. Conclusion and Evaluation

In this study, a 48 W closed-loop buck converter that steps down a 50 V input voltage to a 24 V / 2 A output was designed and simulated in the PSIM environment. The theoretical calculations for the duty cycle, inductance value, and current ripple were found to be in full agreement with the simulation outputs. The operation in Continuous Conduction Mode (CCM) was successfully verified, as the minimum inductor current remained strictly above zero.

Through the implementation of the closed-loop control architecture and Type-III compensation, the output voltage was stably regulated at the 24 V level, and the system's stability was maintained against load transients. Furthermore, the current stresses on the capacitors and semiconductor components were observed to be consistent with the analytical designs. The results demonstrate that the proposed buck converter provides a consistent, reliable, and applicable structure from both theoretical and simulation perspectives.

10. References

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